

Week 5 – Advanced Data Manipulation with Pandas

Customer Sales Analysis

1. Project Overview

This project analyzes customer and sales data using Python and Pandas.

The main objectives are to analyze customer purchasing patterns, sales performance, top customers, products, regions, and customer churn.

2. Objectives

- Load and explore customer and sales datasets.
- Clean and prepare the data.
- Perform grouping and aggregation.
- Apply filtering and string operations.
- Perform date manipulation.
- Use merging and joining where applicable.
- Create pivot tables for data summarization.
- Analyze customer churn.
- Generate useful business insights.

3. Datasets

The project uses two datasets:

- `sales\_data.csv` – Contains sales transaction information.
- `customer\_churn.csv` – Contains customer and churn information.

4. Technologies Used

- Python
- Pandas
- Matplotlib
- Visual Studio Code

5. Data Analysis

The following Pandas operations were performed:

- Data loading using `read\_csv()`
- Dataset exploration
- Missing-value checking
- Duplicate checking

- Data filtering
- Grouping and aggregation
- Date manipulation
- Pivot-table analysis
- Customer churn analysis

6. Sales Analysis

Sales data was analyzed by:

- Product
- Region
- Month
- Quantity
- Total sales

The analysis helps identify high-performing products and regions and understand sales trends.

7. Customer Analysis

The customer dataset was analyzed to understand:

- Customer tenure
- Monthly charges
- Total charges
- Contract types
- Payment methods
- Churn

Churn was also compared across different customer segments.

8. Data Merging

The customer and sales datasets were checked for a common customer identifier.

The available customer ID formats were different, so an unreliable merge was avoided.

9. Pivot Tables

Pivot tables were used to summarize sales data by categories such as:

- Product
- Region
- Month

This makes the data easier to compare and understand.

10. Visualizations

```
File Edit Selection View Go Run ... Week1_Task Update
customer_churn.csv U X main.py U X
Task_5_Advanced_DataManipulation > main.py > main
1 """Week 5 customer and sales analysis using advanced Pandas operations."""

Problems Debug Console Ports Spell Checker 18 Code Reference Log
Terminal
D:\Developer_Arena\Week1_Task\Task_5_Advanced_DataManipulation>python main.py
Customer first 5 rows:
CustomerID Tenure MonthlyCharges TotalCharges Contract PaymentMethod PaperlessBilling SeniorCitizen Churn
C00001 6 64 1540 One year Credit Card No 1 0
C00002 21 113 1753 Month-to-month Electronic Check Yes 1 0
C00003 27 31 1455 Two year Credit Card No 1 0
C00004 53 29 7150 Month-to-month Electronic Check No 1 0
C00005 16 185 1023 One year Electronic Check No 1 0

Sales first 5 rows:
Date Product Quantity Price Customer_ID Region Total_Sales
2024-01-01 Phone 7 37300 CUST001 East 261100
2024-01-02 Headphones 4 15406 CUST002 North 61624
2024-01-03 Phone 2 21746 CUST003 West 43492
2024-01-04 Headphones 1 30895 CUST004 East 30895
2024-01-05 Laptop 8 39835 CUST005 North 318680

Customer dataset shape: (500, 9)
Sales dataset shape: (100, 7)
=====
WEEK 5 - CUSTOMER SALES ANALYSIS
=====

File Edit Selection View Go Run ... Week1_Task Update
customer_churn.csv U X main.py U X
Task_5_Advanced_DataManipulation > main.py > main
1 """Week 5 customer and sales analysis using advanced Pandas operations."""

Problems Debug Console Ports Spell Checker 10 Code Reference Log
Terminal
=====
WEEK 5 - CUSTOMER SALES ANALYSIS
=====
Customer dataset shape: (500, 9)
Sales working shape after derived date columns: (100, 12)
Customer columns: ['CustomerID', 'Tenure', 'MonthlyCharges', 'TotalCharges', 'Contract', 'PaymentMethod', 'PaperlessBilling', 'SeniorCitizen', 'Churn']
Sales columns: ['Date', 'Product', 'Quantity', 'Price', 'Customer_ID', 'Region', 'Total_Sales']
Customer missing values: 0
Customer duplicate rows: 0
Sales missing values: 0
Sales duplicate rows: 0

Sales aggregations:
Total sales by product:
Product
Laptop 3889210
Tablet 2884340
Phone 2859394
Headphones 1384033
Monitor 1348071

Total sales by region:
```

```
File Edit Selection View Go Run ... Week1_Task Update
customer_churn.csv U X main.py U X
Task_5_Advanced_DataManipulation > main.py > main
1 """Week 5 customer and sales analysis using advanced Pandas operations."""

Terminal
Total sales by region:
Region
North 3983635
South 3737852
East 2519639
West 2123922

Monthly total sales:
Month
2024-01 4120524
2024-02 2656050
2024-03 4485006
2024-04 1103468

Total sales: 12,365,048.00
Total quantity: 478

Filtered high-value North/South sales: 14
Filtered long-tenure, high-charge customers: 194

Customer ID overlap: 0
Merge result: no customer-sales merge performed because the ID formats do not overlap.
```

```
File Edit Selection View Go Run ... Week1_Task Update
customer_churn.csv U X main.py U X
Task_5_Advanced_DataManipulation > main.py > main
1 """Week 5 customer and sales analysis using advanced Pandas operations."""

Terminal
Pivot: sales by product and region
Region East North South West
Product
Headphones 288361 107091 512168 476413
Laptop 221946 1798206 1373120 495938
Monitor 642870 397100 39924 268177
Phone 506828 489284 1471428 391854
Tablet 859634 1191954 341212 491540

Pivot: monthly sales by product
Product Headphones Laptop Monitor Phone Tablet
Month
2024-01 241498 1981281 67662 937670 892413
2024-02 73396 285064 772321 838792 686477
2024-03 687734 1522863 477371 1082932 714106
2024-04 381405 100002 30717 0 591344

Churn analysis:
Overall churn rate: 10.60%
Contract
Month-to-month 20.59
Two year 6.94
```

```
Task_5_Advanced_DataManipulation > main.py > main
1 """Week 5 customer and sales analysis using advanced Pandas operations."""

Churn analysis:
Overall churn rate: 10.60%
Contract
Month-to-month    20.59
Two year          6.94
One year          4.30

Churn by payment method:
PaymentMethod    Churn rate
Credit Card      13.48
Electronic Check  11.04
Bank Transfer     6.92

Top customer analysis: unavailable because there are zero matching customer IDs.

Charts saved:
- visualizations\sales_by_product.png
- visualizations\sales_by_region.png
- visualizations\monthly_sales_trend.png
- visualizations\churn_by_contract.png
- visualizations\sales_distribution.png

D:\Developer_Arena\Week1_Task\Task_5_Advanced_DataManipulation>
```

11. Key Insights

- Laptop was one of the highest-performing products in the sales dataset.
- North was the highest-performing region.
- Sales varied across the analyzed months.
- Customer churn was analyzed based on different customer characteristics.
- Month-to-month customers showed higher churn compared with longer-term contracts.

12. Error Handling and Validation

The project includes basic validation for:

- Missing files
- Missing values
- Duplicate records
- Invalid data
- Dataset structure

13. Testing

The project was tested by running:

```
``bash
```

```
python main.py
```

The analysis and visualizations were successfully generated.

14. Conclusion

This project helped demonstrate advanced Pandas operations including data cleaning, filtering, grouping, aggregation, date manipulation, pivot tables, and customer analysis.

The analysis provides useful insights into sales performance and customer churn.

15. Project Structure

Task_5_Advanced_DataManipulation/

```
|
├── main.py
├── sales_data.csv
├── customer_churn.csv
├── requirements.txt
├── data/
├── visualizations/
├── screenshots/
└── report/
    └── analysis_report.md
```